

Distribution of antibiotic and metal resistance genes in two glaciers of North Sikkim, India

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ABSTRACT

Glacier studies as of late have ruffled many eyeballs, exploring this frigid ecology to understand the impact of climate change. Mapquesting the glaciers led to the discovery of concealed world of “psychrophiles” harboring in it. In the present study, the antibiotic resistance genes (ARGs) and heavy metal resistance genes (MRGs) were evaluated through both the culture-dependent and culture-independent methods. Samples were collected from two different glaciers, i.e., debris-covered glacier (Changme Khangpu) and debris-free glacier (Changme Khang). Functional metagenomics of both the glacier samples, provided evidence of presence of resistant genes against various antibiotic groups. Bacitracin resistant gene (*bacA*) was the predominant ARG in both the glaciers. MRGs in both the glacier samples were diversified as the genes detected were resistant against various heavy metals such as arsenic, tungsten, mercury, zinc, chromium, copper, cobalt, and iron. Unique MRGs identified from Changme Khangpu glacier were resistant to copper (*cutA*, *cutE*, *cutC*, *cutF*, *cueR*, *copC*, and *copB*) and chromium (*yefI*, *ruvB*, *nfsA*, *chrR*, and *chrA*) whereas, from Changme Khang glacier they showed resistance against cobalt (*mgtA*, *dmeF*, *corD*, *corC*, *corB*, and *cnrA*), and iron (*yefD*, *yefC*, *yefB*, and *yefA*) heavy metals. ARGs aligned maximum identity with Gram-negative psychrotolerant bacteria. The cultured bacterial isolates showed tolerance to high concentrations of tested heavy metal solutions. Interestingly, some of the antibiotic resistant bacterial isolates also showed tolerance towards the higher concentrations of heavy metals. Thus, an introspection of the hypothesis of co-occurrence and/co-selection of ARGs and MRGs in such environments has been highlighted here.

1. Introduction

Earlier decades have witnessed an intensified prototype dissemination of antibiotic-resistant microorganisms and these antibiotic resistant microbes surmounts a great threat to human health worldwide (Taccaroni et al., 2018). Although the antibiotic resistance occurs naturally, however human activities including misuse and overuse of antibiotics have exacerbated the situation. Antibiotic resistance is an ancient phenomenon and have been reported from pristine environments such as 30,000 year old permafrost sediments (D’Costa et al., 2011) and ancient arctic sediments (Perron et al., 2015); which are supposed to be free from pharmaceutical or chemical driven antibiotic resistance. However, the antibiotic resistance genes (ARGs) from these pristine environments are different from pharmaceutical driven ARGs. Environmental microbes carrying the ARGs are also capable of transferring or inducing these genes to clinically important pathogens (Rizzo et al., 2013).

Antibiotic resistance phenomenon in pathogenic and non-pathogenic bacteria belonging to the mesophilic world has been extensively studied; however, very few attentions have been paid towards antibiotic resistance in psychrophilic bacteria. The distribution of ARGs and their antibiotic resistance profile needs attention to avert the possible emergence of an opportunistic pathogen of a psychrophilic origin.

A new perspective of human health have aroused from antibiotic resistant bacteria harboring in glacial ecology. Many ARGs for antibiotics like aminoglycosides, lactams etc. were reported from glacier ecosystem of snow, ice, permafrost regions etc. (Edwards, 2015; Segawa et al., 2013). Functional analogues and structural homologues of various beta-lactamases were detected from ancient Alaskan soils by COG and KEGG analysis (Allen et al., 2009). The most probabilistic hypothesis put forward is due to the air borne bacteria carried as in aerosols or in dust particles might help in the dispersal of these ARGs whereas some hypothesized them to be from migratory bird droppings (Segawa et al.,

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2013). Dancer and his co-workers also reported many ampicillin resistant coliforms in the 2000 years old glacial ice core samples from Canadian Arctic archipelago (Dancer et al., 1997). The presence of ARGs in environment samples represents the anthropogenic closeness of human population which also might lead to transfer of genes through glacial melt water in future. Also, heavy metal resistance genes (MRGs) can get transferred owing to inorganic pollutant contamination (Edwards, 2015; Hodson, 2014; Tieber et al., 2009).

Wide distribution of heavy metals in nature is also another facet of industrialization and urbanization which has negatively impacted human health (Iyer et al., 2005). Microorganisms have incorporated the heavy metal resistance as a selective evolutionary criterion. Studies widely report the coexistence of ARGs and MRGs in bacteria (Cesare et al., 2016). Cross-resistance and presence of ARGs and MRGs in mobile genetic elements with horizontal transfer efficiency is the underlying mechanism of co-existence. These inquisitive properties and paradigm balancing act between ARGs and MRGs and their co-existence among the psychrophilic bacteria needs to be introspected and investigated. Hence, our investigation is the first of its kind to be reported from the Sikkim Himalayan glaciers.

The Himalayas alias “The Third Pole of the World” is the origin and source of global freshwater resources to which nearly 5.4 billion populations depend. They are the largest depository account of snow cover, cryosphere, and inhabit numerous glaciers apart from the Polar Regions (Bajracharya and Shrestha, 2011). The Himalayas are the epitome of cold world where both macro and micro level of biodiversity allows many species to enrich its vast geography. Water is the most important component of a biosphere which determines the species richness as well as it ensures the human biogeography elements nourishment. Sikkim, a North East Indian state, lies in a landlocked zone sandwiched between Bhutan, Nepal and China. Thus, it is a very unique hot spot zone in the Eastern Himalayas and sometime due to its much diversified ecology and topography, it is also referred to as Sikkim Himalayas. This part of the Eastern Himalayas plays an important role on the livelihood of the millions of people and limited studies on microbiology of their inhabiting glaciers have been conducted with the aim of identifying few novel species (Kumar et al., 2015a, b; 2016). Thus, till date there has been no study on the metal and antibiotic resistance bacteriome of Sikkim Himalayas. The aim of our present study was to explore the ARGs and MRGs resistance among the psychrophilic bacteria isolated from the accumulation zone of the two glaciers of North Sikkim as well as to investigate their possible co-existence if any present among the glacial bacteriome.

2. Materials and methods

2.1. Description of sampling sites

The accumulation zone of Changme Khang and Changme Khangpu glacier were selected for the current study located at North district of Sikkim. The debris-covered Changme Khangpu glacier (CK) and debris-free Changme Khang glacier (CKG) are located at the upper catchment of Teesta river basin, in the Sebu valley of North Sikkim, India. Both the glacier originated from the south slope of Gurudongmar peak, North Sikkim. The geographical position of coordinates and elevation range of Changme Khang and Changme Khangpu glacier were measured with the help of GPS MAP 78S (Germin, India), an automated Global Positioning device. The map of the glacier sites was prepared with the help of Google Earth Software. Changme Khangpu glacier is located at an altitude of 5221.2 m, latitude 27°58′04.16″N; longitude 88°42′56.68″E whereas Changme Khang glacier is positioned at an altitude of 5093.8 m, latitude 27°56′38.80″N, and longitude 88°39′56.91″ E (Fig. 1).

In debris-free glacier, the accumulation zone is not covered by the deposition of rocks and boulders from the surrounding mountains and glaciers remains crystal clear. On the other hand, debris-covered glaciers are mainly responsible for sediment transport in glacial environments (Kirkbride, 2000). The origin of the Sikkim Himalayan debris-covered and rock glaciers is still under discussion. Debris caused due to the geomorphological processes on the sedimentary systems is well known (Keiler et al., 2010). Based on records of past events, geomorphological hazards such as landslides, debris flows, mudflows and other expressions of slope instability can result from climate variability or anthropogenicity or in combination (Basnett et al., 2013; Garg et al., 2019; Keiler et al., 2010; Nijampurkar, 1985; Racoviteanu et al., 2015; Raj et al., 2013; Shukla et al. 1983, 2018; Tanarro et al., 2019). However, the knowledge about regional snow line fluctuations and rate of debris cover increase is also limited. The possible influence of important factors such as glacier size, debris cover, and glacial lake on the multidimensional glacier response (i.e., a variation of other glacier parameters) also remains poorly constrained (Garg et al., 2017a). Thus, to have a clear picture about the status and behavior of the Sikkim Himalayan glaciers, a multi-parametric monitoring and assessment is required (Garg et al., 2017b).

2.2. Ice core sampling and pretreatment

The ice core samples were taken by drilling about 2 m depth as per Zhang et al. (2003) and the glacier ice core samples was collected from

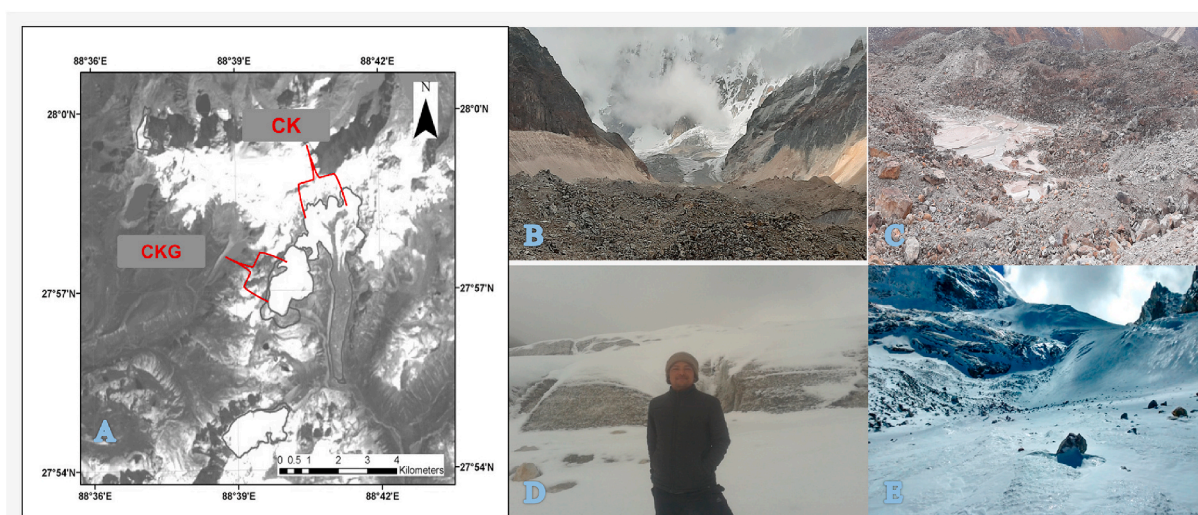


Fig. 1. Google Earth map of Changme Khang (CKG), and Changme Khangpu (CK) of North Sikkim.

different locations within the accumulation zone from Changme Khang (CKG) and Changme Khangpu (CK) glacier, North Sikkim, India. The samples were immediately transported to the laboratory in sterile ice thermal containers maintaining the geomorphology of the glacier samples at 4°C (Zhang et al., 2003). The glacier ice core sample was processed and cut into small pieces about six inches. The aseptic measures were taken and the glacier ice core samples were analyzed as per Zhang et al. (2003). The ice core samples were cut with a sterilized saw-tooth knife and around 5 mm annulus was discarded. The remaining inner ice core was rinsed with cold ethanol (4°C; 95%), and finally with cold (4°C) autoclaved water. The glacier ice core samples were placed in the sterile glass beaker (Borosil, India) and melted at 4°C incubator. These handling processes were performed at temperatures below 16°C aseptically using positive pressure laminar air flow.

2.3. Isolation and characterization of the isolates

A total of 17 different bacterial isolates were isolated and their identification was done earlier by Sherpa et al. (2018a). For further characterization, heavy metal resistance and antibiotic sensitivity of the culturable isolates were analyzed. As, there have been very few studies on metallo-antibiotic resistant bacteria from the glacier, hence this study is a significance one.

2.4. Antibiotic susceptibility studies

Antibiogram of the selected isolated from Changme Khang and Changme Khangpu glacier samples in this study were done on Mueller Hinton Agar (MHA) plates using the Kirby Bauer Antibiotic Sensitivity method (Bauer et al., 1999). Overnight culture broth (0.1 mL) was swabbed by sterile cotton in Mueller Hinton Agar (MHA) plates and sterile antibiotic discs were placed on the swabbed plate. A sterile membrane filter paper discs (0.4 µm) immersed in sterile water was used as negative control. Antibiotic sensitivity was measured against Ampicillin (Amp10µg), Chloramphenicol (C30µg), Streptomycin (S100µg), Vancomycin (Van30µg), Methicillin (Met10µg), Tetracycline (Tet30µg), and Erythromycin (Er15µg). The plates were incubated at 15°C for 72hrs. A clear zone of inhibition was measured accordance with BSAC standardized disc susceptibility testing method (version 6) (Andrews, 2007).

2.5. The minimum inhibitory concentration (MIC) of antibiotics

The MIC assays were performed according to the CLSI guidelines, 2004 (Lyamlouli et al., 2011). Tube dilution method was performed to check the MIC. Tubes containing 10 mL of Luria Bertani broth and various antibiotics at different concentrations (between 0.5 and 8 mgmL⁻¹ using dilution factors of ½) were inoculated with 200 µL of 72hrs old culture (O.D. at 600 nm=>1) of each isolated bacterial strain. A negative control consisted of Luria Bertani broth without antibiotics. Tubes were examined in uv/vis spectrophotometer (uv-vis 1600-00BD PerkinElmer LAMBDA 40, USA) at 600 nm after incubation at 15°C for 72 h.

2.6. Heavy metal tolerance to isolates

2.6.1. Screening and assessment of metal toxicity

A tube dilution method was performed to evaluate the MIC (Hassen et al., 1998). Culture medium and metallic solution were autoclaved. 0.5 mL of appropriate metals concentration was dissolved in Antarctic Bacterial broth (AB broth) with a final volume of 10 mL. Three tubes were prepared for each metal concentration, and then inoculated with 200 µL of a 72h old culture of the studied bacterial strains. A positive control consisted of AB broth inoculate with the bacterial isolate and a negative control consisted of a metal supplemented medium without adding the bacterial isolate. To constitute the metal supplemented

medium 1 mL of respective metal solution was added to 9 mL of AB broth. The glassware used for the experiments was leached in 2N HNO₃ and then was rinsed thrice with distilled water (MilliQ) so as to rule out any possibilities of heavy metal contamination. Tubes were read in uv/vis spectrophotometer (PerkinElmer LAMBDA 40) at 600 nm after incubation at 15°C for 72 h (Kim et al., 2007).

2.7. Metagenomic DNA extraction

Glacier ice community genomic DNA was extracted by using DNeasy Power water kits (MoBio Laboratories, Carlsbad, CA, USA) as described in the manufacturer's instructions. Quality of the DNA was checked on 0.8% Agarose gel and the DNA was quantified using Qubit fluorimeter (ThermoFisher, USA), with the detection limit of 10–100 ngµL⁻¹. The shotgun sequencing libraries were prepared using NEB Next Ultra II DNA library preparation kit for Illumina (New England BioLabs, UK) according to the manufacturer's instructions. The fragment size distribution of each library was analyzed by Bioanalyzer and High Sensitivity DNA Kit (Agilent, Germany). Each library was sequenced using 100bp paired-end, HiSeq 2000, Illumina platform after its quantification by qPCR.

2.8. Metagenomic data processing and analysis

The 18.72 million reads were optimally assembled using the metaSPAdes assembler v3.13 (Nurk et al., 2017), and the optimal assemblies were performed with *k*-mer length of 20 in increments of 20 and also scaffolded. Out of the 5 assemblies (i.e. 41, 61, 81, 101 *k*-mer and scaffold) it was concluded based on N50, the number of contigs assembled and contig length distribution that scaffolded assembly performed the best and was taken for further analysis. The assembly statistics were calculated using QUAST software and metaSPAdes after scaffolding assembled into contig length greater than 500bp (Gurevich et al., 2013). The reads from the sample were mapped onto the contigs using BOWTIE2 in order to check for the assembly quality. More than 60% of sequence reads mapped back to the contigs successfully (Langmead and Salzberg, 2012).

2.9. Functional analysis of the metagenome data

The assembled contigs were annotated using BLASTn against the NCBI-nucleotide custom metagenome database that included all nucleotide entries for organisms that belong to the archaea, bacteria, fungi and viruses; with minimum sequence similarity of 60% and e-value lesser than 1e-07 (Altschul et al., 1990). The contigs were annotated with ~80% contigs with similarity >90%. The gene prediction and annotation of the assembled contigs were carried out using PROKKA (Seemann, 2014). The predicted genes were annotated further for KEGG orthology using the KEGG Automatic Annotation Server (KAAS) (Moriya et al., 2007), and Cluster of Orthologous Groups (COG) by BLASTx against the COG database.

2.10. Antibiotic resistance genes (ARGs) analysis

For the annotation of ARGs, the metagenomic tags of each sample were blasted against the structured non-redundant clear antibiotic resistance gene database (ARBD, <http://arbd.cbcb.umd.edu>) (Liu and Pop, 2009). The sequence is considered as ARG like sequences according to its best BLAST hit in ARBD with a threshold of amino acid sequences identity of not less than 90% to the references sequences and had a query coverage of not less than 25 amino acids.

2.11. Metal resistance genes (MRGs) analysis

For annotation of MRGs, metagenomic data of each MRG genes were BLAST matched against the structured non-redundant clear metal

resistance genes by searching on online BacMet database (<http://bacmet.biomedicine.gu.se>) (Pal et al., 2014). The sequences were considered to be MRGs following the same cutoff; similarity not less than 90% to the references sequences and had query coverage of not less than 25 amino acids. The Changme Khang and Changme Khangpu glacier metagenomic reads were to Sequence Read Archive (SRA), NCBI to obtain the Bio-Sample and accession number. The accession numbers for Changme Khang glacier are SAMN08732116, SRP136080 and SRS3073134 with sample name as Changme Khang glacier on the other hand the accession numbers for Changme Khangpu glacier are SAMN08741619, SRP136079, and SRS3073133 with sample name as CKVV4.

2.12. Statistical analysis

Assessment of the heavy metal tolerance of two glaciers isolates (Changme Khangpu and Changme Khang glacier) on the basis of MIC was carried out by Principal Component Analysis (PCA). It was done with the help of XLSTST 2014.03 software.

3. Results

3.1. Antibiotic sensitivity of glacier bacterial isolates

Previously isolated 17 different bacterial species, from Changme Khang and Changme Khangpu glacier ice core samples were used for this study (Table 1a and 1b) (Sherpa et al., 2018a). Further characterization of antibiotic sensitivity of the culture isolates were performed on MHA plate by disk diffusion method. Six different antibiotics such as vancomycin, erythromycin, ampicillin, streptomycin, chloramphenicol, and methicillin were used. These results were compared with the zone size of Gram-negative and Gram-positive bacteria such as *Enterobacter* and *Staphylococcus* as per CLSI guidelines. It suggested that the majority of the isolates (72%) showed resistance to ampicillin, and methicillin (63%) antibiotics as shown in (Table 1a and 1b). The results have shown

Table 1

Antibiotic Susceptibility of Changme Khangpu glacier isolates. S=Sensitive; I=Intermediate; R = Resistant; Van = Vancomycin; Er=Erythromycin; Amp = Ampicillin; S100 = Streptomycin; C=Chloramphenicol; Met = Methicillin.

Antibiotic disc with their concentration in mcg Zone of inhibition (in mm)							
Isolates	Van30	Er15	Amp10	S100	C30	Met10	MAR
CK1	30; S	40; S	19; S	0; R	40; S	0; R	0.33
CK3	36; S	40; S	14; S	37; S	33; S	40; S	0
CK5	40; S	40; S	0; R	39; S	40; S	0; R	0.33
CK6	28; S	36; S	0; R	40; S	40; S	12; I	0.16
CK9	30; S	35; S	10; R	32; S	40; S	0; R	0.33
CK10	32; S	32; S	14; S	38; S	0; R	25; S	0.16
CK11	36; S	18; I	10; R	16; R	16; R	25; S	0.5
CK13	29; S	0; R	10; R	40; S	40; S	0; R	0.5
CK15	28; S	30; S	10; R	39; S	25; S	26; S	0.16
CK16	29; S	40; S	22; S	40; S	40; S	0; R	0.16
CK17	12; R	20; S	21; S	40; S	40; S	0; R	0.33
CK19	29; S	40; S	21; S	0; R	40; S	0; R	0.33
CK20	25; S	35; S	0; R	35; S	36; S	0; R	0.33
CK21	20; S	40; S	0; R	0; R	40; S	0; R	0.5
CK22	15; S	24; S	10; R	30; S	30; S	12; I	0.16
Antibiotic disc with their Concentration in mcg Zone of inhibition (in mm)							
Isolates	Van30	Er15	Amp10	S100	C30	Met10	MAR
CKG1	30; S	40; S	0; R	22; S	40; S	17; R	0.33
CKG2	29; S	40; S	0; R	31; S	40; S	0; R	0.33
CKG4	38; S	40; S	11; R	40; S	40; S	35; S	0.16
CKG5	29; S	40; S	0; R	40; S	40; S	0; R	0.33
CKG6	40; S	40; S	0; R	40; S	40; S	28; S	0.16
CKG7	0; R	0; R	0; R	40; S	0; R	0; R	0.83
CKG8	30; S	40; S	0; R	40; S	40; S	0; R	0.33

that the resistance patterns of antibiotics were highest against ampicillin followed by methicillin, streptomycin, chloramphenicol, erythromycin, vancomycin respectively. Similarly, only three isolates showed a multiple antibiotic resistance (MAR) index of 0.5.

3.2. Minimum inhibitory concentration (MIC) of glacier isolates

The four different groups of antibiotics were tested against the glacier isolates for their MIC values. The test antibiotics were Beta-lactam (ampicillin), phenicol (chloramphenicol), aminoglycoside (streptomycin), and macrolide (erythromycin). Various dilutions from stock solution were made as 0.25, 0.5, 1, 2, 4, 8, and 16 mgL⁻¹. The highest MIC value shown by the isolates against ampicillin, chloramphenicol, and erythromycin antibiotics were 4 µg L⁻¹, 8 µg L⁻¹ and 8 µg L⁻¹ (Fig. 2). Till-date there is not a single globally accepted guideline for susceptibility testing or breakpoint testing for psychrophilic/psychrotolerant bacteria. However, psychrotolerant bacteria such as *Pseudomonas* sp. and *Enterobacter* sp. and their breakpoint as per CLSI, 27-edition data as well as two *Bacillus* species were taken as baselines to investigated the resistance pattern of our isolates (Rosenquist et al., 2005).

3.3. Metagenomic studies of antibiotic resistance genes (ARGs)

Metagenomic sequence data was generated using Illumina HiSeq and study of ARGs were carried out using arbAnno V.1.0. For functional metagenomic analysis of two glaciers, i.e., Changme Khang (debris-free) and Changme Khangpu (debris-covered) glacier were selected. The results have shown the different resistance genes in both the glaciers. Overall higher numbers of resistance genes were detected in Changme Khangpu glacier. In both the glacier, the most dominant resistance genes were *bacA* (bacitracin), whereas *mexD* genes (multidrug resistance) were present only in Changme Khang glacier. The higher percentages of tetracycline resistance gene such as *tet41* (4.03%) from Changme Khangpu glacier and *tetC* (14.6%) were detected from Changme Khang glacier whereas aminoglycoside resistance genes and their numbers were similar in both the glaciers. The aminoglycoside gene was showing highest similarity (99%) with *Serratia marcescens* (*aac6ic*) gene, *Acinetobacter baumannii* (*aph3ia*) gene, and *Escherichia coli* (*KsgA*) gene. Similarly, bacitracin gene was showing 99% homology with *Bacillus cereus* (*bacA*) gene, and Quinolone resistance gene with *Serratia proteamaculans* (*qnrB*) gene. The tetracycline resistance gene showed 99% identity with *Serratia marcescens* (*tet41*) and *Bifidobacterium bifidum* (*tetC*) gene whereas multidrug resistance genes demonstrate highest similarity with *Serratia proteamaculans* (*qnrB*) and (*rosB*) gene, *Pseudomonas putida* (*mexD*) gene and *Escherichia coli* (*mdtG*) gene (Fig. 3a and b).

3.4. Heavy metal tolerance of glacier isolates

3.4.1. Screening and assessment of metal toxicity

Five different metals such as FeCl₂, ZnCl₂, CoCl₂, NiCl₂, CuCl₂, and HgCl₂ were screened and assessed for metal toxicity by tube dilution method to check MIC. An experiment in liquid culture broth was conducted to determine the precise concentrations of metals at which the isolates could grow. The mercury was highly toxic to almost all the strains as only less concentration <0.1 mM of HgCl₂ was tolerable. In terms of minimum inhibitory concentrations against mercury, seventeen isolates showed the MIC value of 1 mM and four isolates showed MIC value of 0.5 mM. A total of nine isolates exhibited MIC value against zinc was 3 mM followed by each four isolates showing MIC values zinc of 5 mM, 2 mM, and 1 mM respectively. The MIC values of CuSO₄·5H₂O revealed by the isolates were 3 mM (ten isolates), 1 mM (eight isolates) and 0.5 mM (two isolates). The MIC value of isolates showed against nickel was 3 mM (fifteen isolates) and 1 mM (six isolates). The eighteen isolates showing MIC value of cobalt at a concentration of 1 mM and only three isolates demonstrated MIC value 3 mM (Supplementary

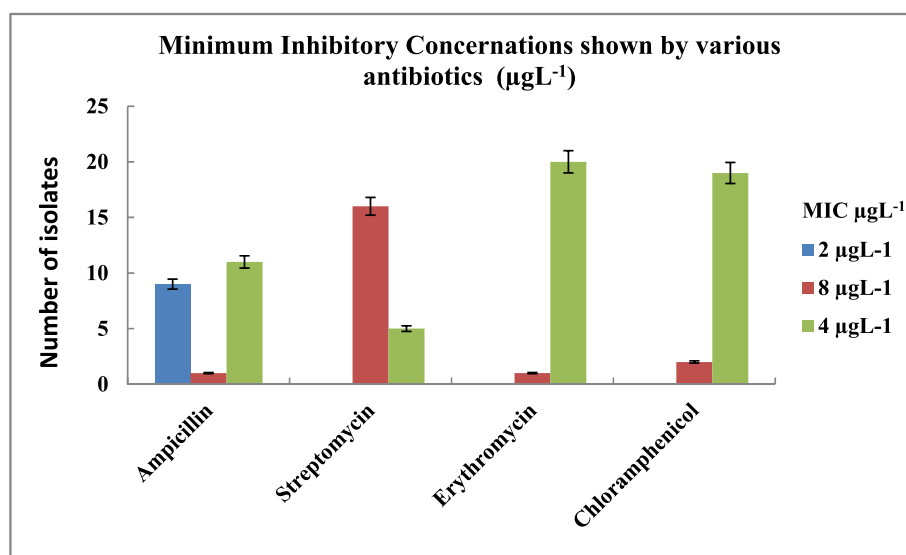


Fig. 2. Minimum inhibitory concentration is shown by various antibiotics in $\mu\text{g L}^{-1}$.

Table 2a).

Our main aim to check the metal tolerance was in order to discover some psychrotolerant metal tolerant and/resistant bacterial isolates that can be employed in bioremediation and also verify the co-occurrence of heavy metal resistance and antibiotic resistance. Interestingly, some of our isolates showed tolerance towards heavy metals such as zinc chloride, which had highest MIC value of 5 mM, followed by copper sulfate, nickel chloride, and cobalt chloride whose MIC values were 3 mM each respectively and for mercury chloride it was 1 mM. As compared with the heavy metals check point (ZnCl_2 , CuCl_2 , NiCl_2 , CoCl_2 (1 mM each) and HgCl_2 (0.5 mM) as described by Nies (1999) our isolates showed higher MIC values (Fig. 4a and b).

3.5. Metagenomic studies of metal resistance genes

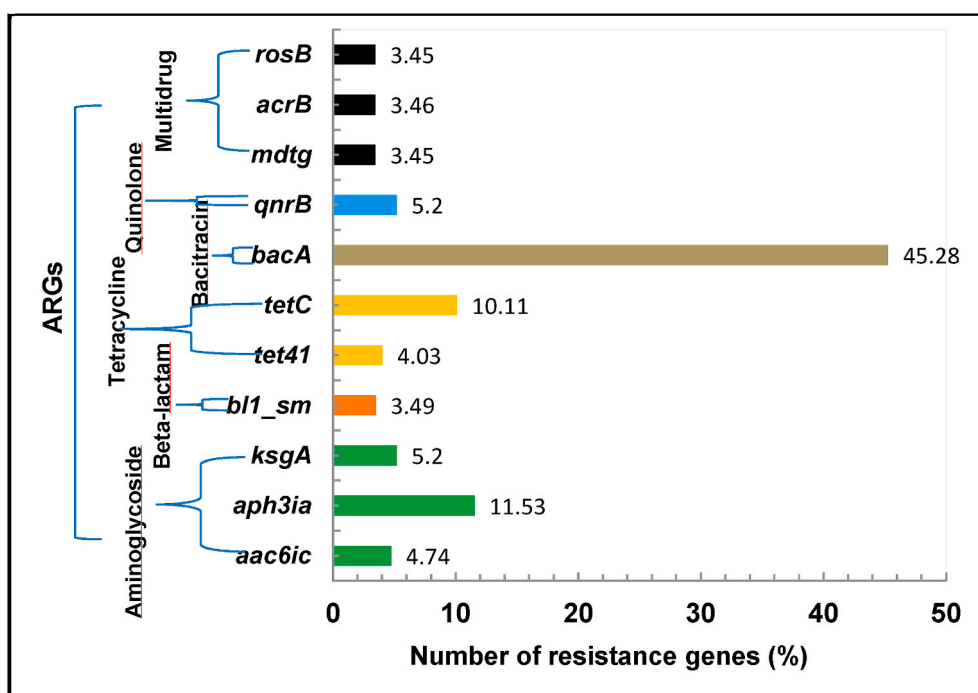
Metagenomic study of the MRGs were carried out using BacMet Scan V.1.0. The results have shown the different resistant genes in both the glaciers. The resistance genes detected, belonged to various metals such as mercury, copper, chromium, and arsenic. The mercury, copper and chromium resistance gene were found in both the glacier except arsenic resistance gene was detected in Changme Khangpu glacier only. The copper resistance genes (*copC* and *cut F*) showed 96% homology *Pseudomonas fluorescens* and *Escherichia coli* K12. The mercury resistance genes (*merA*) showed 98% homology with *Thiobacillus ferrooxidans*, *Staphylococcus aureus*, *Pseudomonas stutzeri*, and *Pseudomonas aeruginosa*. The chromium resistance gene (*chrA*) showed homology with *Pseudomonas aeruginosa*. However, arsenic resistance gene (*arsH*) showed 91% homology with *Yersinia pestis* that was detected only in Changme Khang glacier. Based on the cluster of orthologous groups (COG) classification, various metals resistance genes were envisaged subsequent to each metal resistance classes. However, metal resistances were diverse in Changme Khangpu glacier. The metal resistance was against two different heavy metals, detected in Changme Khang glacier were copper with COG predicted genes such as *cutA*, *cutE*, *cutC*, *cutF*, *cueR*, *copC*, and *copB*, and chromium with COG predicted genes were *yefI*, *ruvB*, *nfsA*, *chrR*, and *chrA* (Fig. 5a). Similarly, from Changme Khangpu glacier COG predicted genes for cobalt were *mgtA*, *dmeF*, *cord*, *corB*, *corC*, and *cnrA* and COG predicted genes for iron were *yefD*, *yefC*, *yefB*, and *yefA* (Fig. 5b).

3.6. Statistical significance of heavy metal tolerance through principal component analysis (PCA)

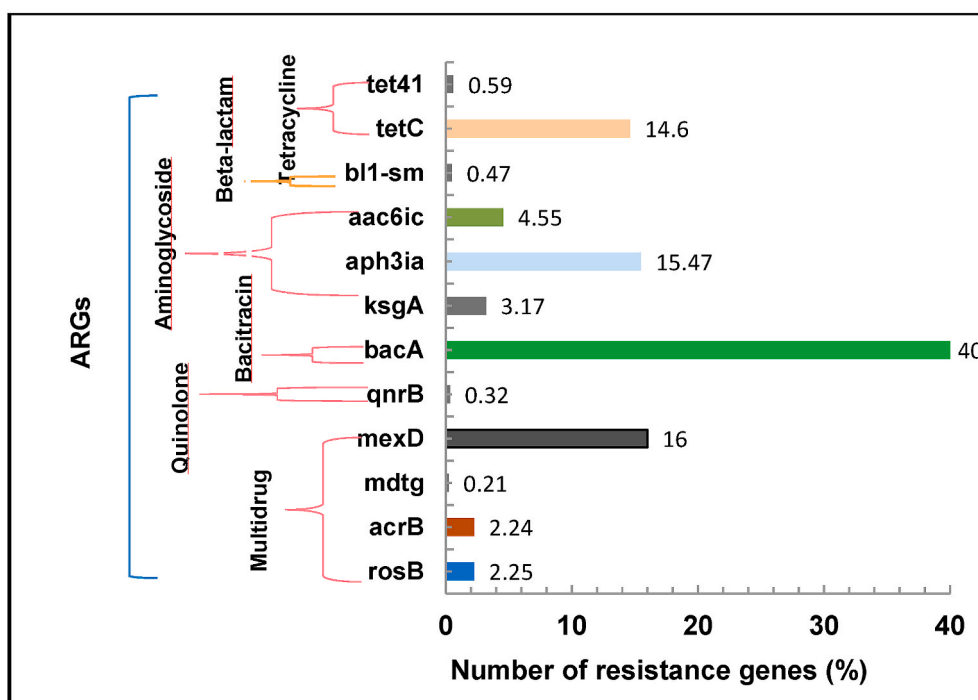
Principal component analysis of heavy metals was calculated and correlated with the studied isolates. The Eigen values among the five principal components were >1 such as 1.876, and 1.210, as shown in (Supplementary Table 2b). The overall variance of the principal component was 37.5%. The Bartlett's sphericity test was also done which signified the test for null-hypothesis that the correlation matrix has an identity relation. Taking this into account, these tests offer the minimum standard to proceed to factor analysis. The results illustrated by these test with the p-value of 0.0092 (threshold value of <0.05), thus represents the factor analysis is valid and considerable as shown in (Supplementary Table 2c). PCA results were comparable to results achieved from MIC analysis i.e., PCA confirms positive correlation among the individual heavy metals to those isolates which were having elevated MIC values with respect to that particular heavy metals. There was a positive correlation between CuSO_4 , ZnCl_2 , and NiCl_2 . However, metals such as CoCl_2 and HgCl_2 were negatively correlated (Fig. 6). The PCA result showed that the highly tolerant strains were present in the vicinity of the concerned heavy metal. The studied isolate such as CK19, CK17, CK10, CK13, CK11, CKG6, and CK16 are present in the vicinity of Cu, Zn, and Ni, which inferred that these isolates were positively correlated with Cu, Zn, and Ni (Fig. 6). From, Changme Khang and Changme Khangpu studied isolates, it can be inferred that the isolates showed the highest tolerance to $\text{Zn} > \text{Ni} > \text{Cu} > \text{Co} > \text{Hg}$ on the basis of MIC results in liquid media.

4. Discussion

Co-selection of antibiotic and metal resistance in the environmental samples might be due to diverse habitats contaminated with various metals present either in their source mineral ores or anthropogenic factors. Co-selection is not restricted to a paradigm of metals, or ecology or microorganisms genetic inheritance through resistivity. It can be simply due to natural genetic capability also. The studies on co-selection present an unambiguous replica for both antibiotic and heavy metal resistivity as impeccable co-selection. Co-resistance occurs when the specific resistant genes are positioned at the same loci on the transferable genetic elements such as a plasmid (Chapman, 2003). This genealogical relationship results in their co-existence. Decade old notions and proofs has established that metal and antibiotic-resistance genes are phylogenetically linked, especially within the mobile genetic plasmids,



a. Antibiotic Resistance genes (ARGs) distribution of Changme Khangpu glacier.

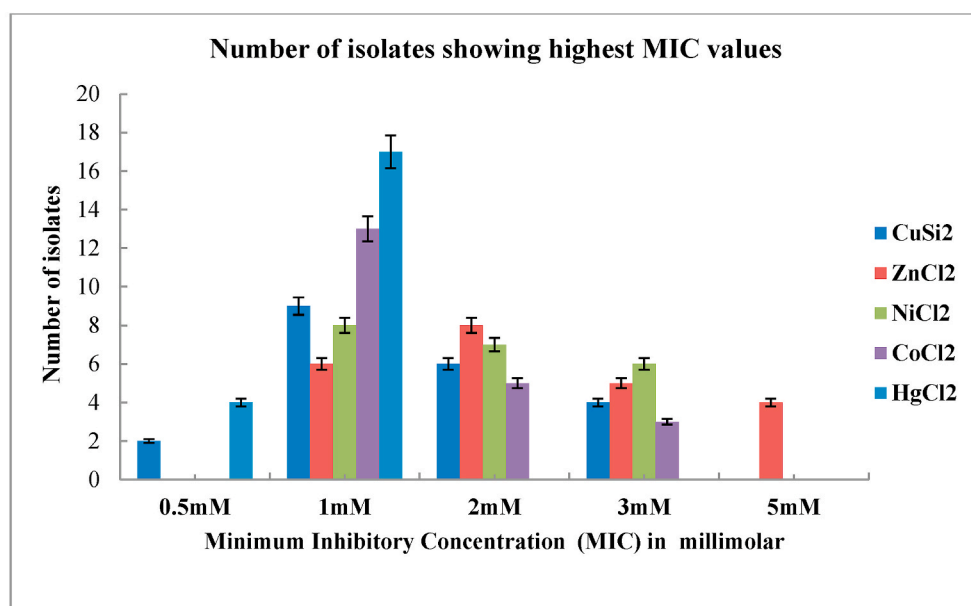


b. Antibiotic Resistance genes (ARGs) distribution of Changme Khang glacier

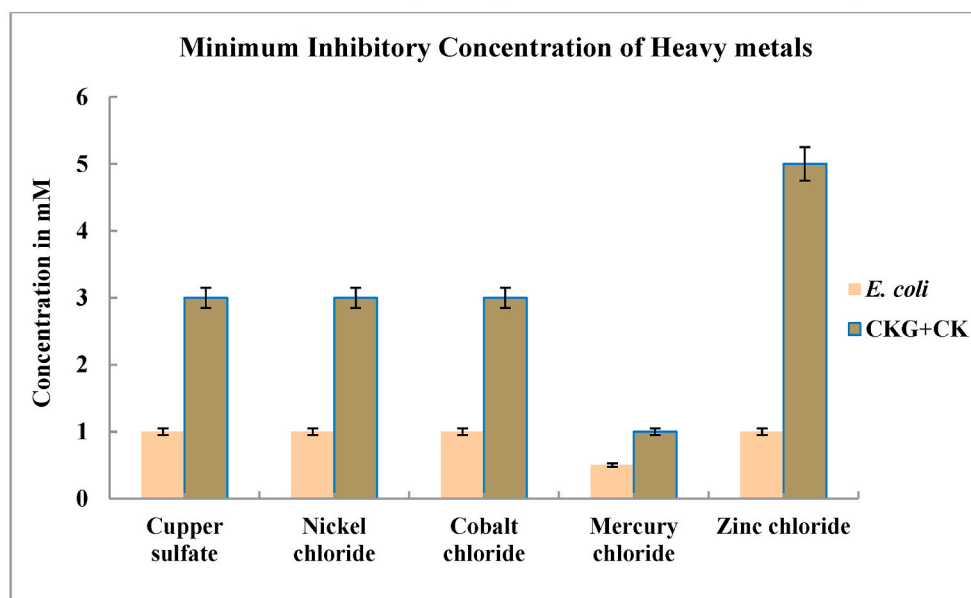
Fig. 3. Antibiotic Resistance genes (ARGs) distribution of Changme Khangpu and Changme Khang glacier.

based on their evidence to transform the other non-resistant microbes or create the potentiality in them to tolerate and resist, proved through functional genomics and gene sequencing studies (Novick and Roth, 1968; Foster, 1983). In case, of glacial environments there are huge possibilities of biofilm formation so as to survive in the extreme conditions through mutualism and synergistic approach. Ultimate goal is to save energy and obtain life support for the microbiota to co-exist in the ecosystem. Biofilms are predominant features in environmental

ecosystems. Bacteria encased within such niches exhibits unusual genotypic functionalities and can easily grow in cryoconite holes, crevices, deep-sub glacial surfaces and require a multilayered, cellular complex matrix-embedded community performing the coordinated expression through co-selective array of genes (Baker-Austin et al., 2006). The glaciers concealed environment acts as natural incubators for the close proximity and similar hierarchy of microbes which also represents an ideal environment for lateral gene transfer. Several



a. The number of isolates showing the highest MIC values for each heavy metal



b. Comparison of minimum inhibitory concentrations among CK+CKG isolates and *E. coli* (Nies, 1999) data.

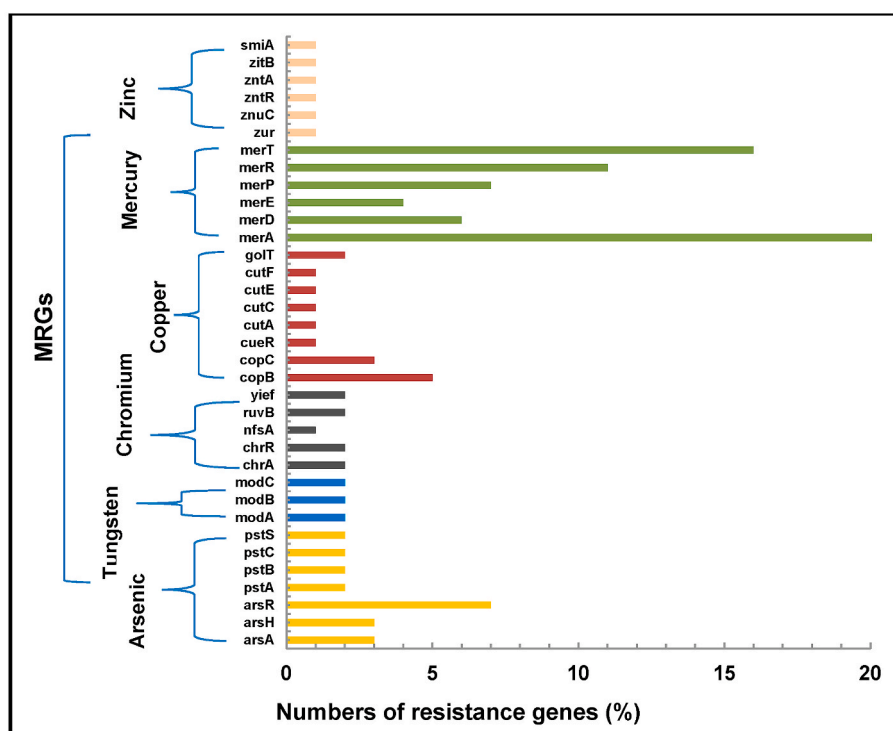
Fig. 4. a. The number of isolates showing the highest MIC values for each heavy metal. b. Comparison of minimum inhibitory concentrations among CK + CKG isolates and *E. coli* (Nies, 1999) data.

discussions on co-existence of metal and antibiotic resistant microbes have been reported in various ecosystem such as rivers, sewage, sludges etc. but the interesting discoveries from pristine and unexplored regions of deep sub surface caves, glaciers and other earth's territory still remains an enigma (Baker-Austin et al., 2006). Thus, natural selection plays a very crucial role for the resistance genes which can only be explained through metagenomic studies and culture dependent studies to correlate and substantiate both the genotype and phenotype characteristics of the microbiota present in them.

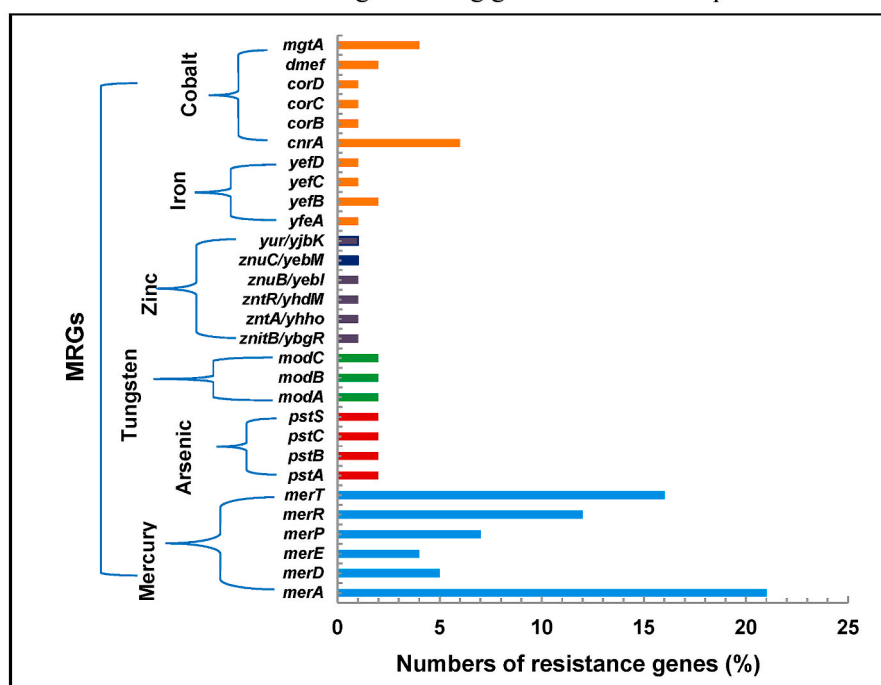
One of the most important aspects of a functional metagenomic approach is the classification of antibiotic and metal resistance-related genes and its functions. An antibiotic-resistant bacteria which can threaten the human race will become a great concern and a large

number of researches have been carried out to curb from this menace throughout the world (Roca et al., 2015). Metagenomic analysis of resistance genes have played potent roles and have become a promising culture-independent method of resolving ARGs and MRGs diversity and abundance from various environments. From the several environments, ARGs and MRGs gene were detected by many researchers such as estuary and deep-sea sediment (Chen et al., 2013), drinking water (Fernando et al., 2016), glacier environments (Segawa et al., 2013).

In the present study, we aimed to check the antibiotic and metal resistance, ARG and MRG genes pattern in two closely located glaciers of Sikkim. The study was carried out using culture-dependent and independent methods. We checked the antibiotic resistance profile of six antibiotics by disc diffusion method such as ampicillin (30mcg),



a. Distribution of MRGs in Changme Khang glacier ice core sample



b. Distribution of MRGs in Changme Khangpu glacier ice core sample

Fig. 5. Distribution of MRGs in Changme Khang and Changme Khangpu glacier ice core sample.

erythromycin (15mcg), chloramphenicol (30mcg), and streptomycin (100mcg), methicillin (10mcg), vancomycin (30mcg) and interestingly found that some of the isolates were resistant against major tested antibiotics. Further, the minimum inhibitory concentrations of these antibiotics against our isolates from Changme Khang and Changme Khangpu glaciers were also checked. In the case of chloramphenicol and erythromycin, the MIC values were $8 \mu\text{gL}^{-1}$ respectively. The least MIC of $4 \mu\text{gL}^{-1}$ was showed against ampicillin and streptomycin. On the basis

of these guidelines, we found that the majority of our isolates were resistant against various tested antibiotics such as ampicillin (72%), methicillin (63%), vancomycin (9%), erythromycin (9%), streptomycin (18%), and chloramphenicol (13%) and also their MIC values were high $8 \mu\text{gL}^{-1}$ (erythromycin, streptomycin and chloramphenicol). Rafiq and his colleagues, studied the Siachen glacier and found that their isolates were resistant to many antibiotics, for example, 64% isolates were resistant to vancomycin and 58% were resistant against methicillin

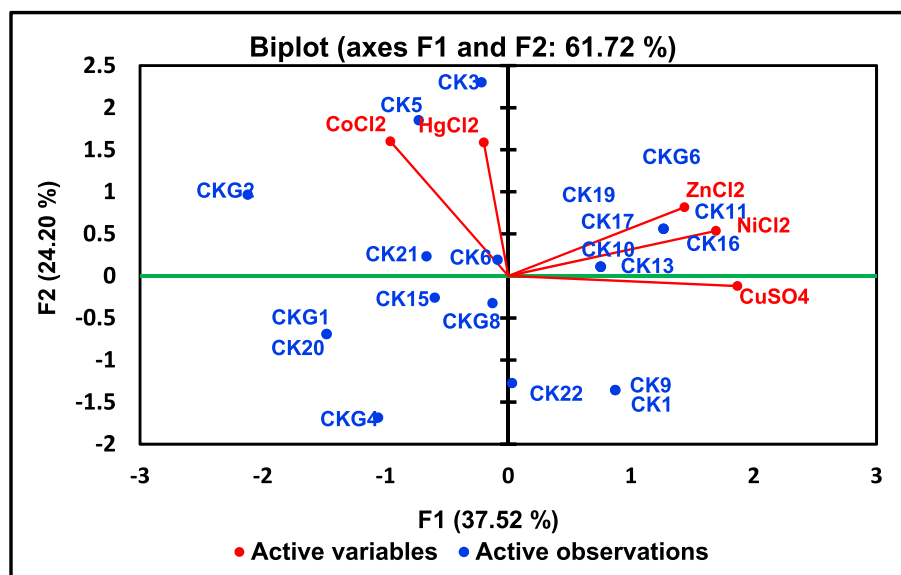


Fig. 6. Principal Component Analysis showing the correlation between heavy metals and various studied isolates.

(Rafiq et al., 2017). Our result such as high number of isolates against methicillin antibiotic correlated with the studies on the Siachen glacier (Rafiq et al., 2017); however other classes of antibiotics showed relatively less resistance. The majority of the studies from pristine environments like ancient Siberian permafrost, alpine glacier cryoconite, and alpine soil suggested that the bacterial isolates from these cold environments were generally sensitive to antibiotics (Gupta et al., 2015; Zhang et al., 2010; Berge and Barnathan, 2005). According to Segawa et al. (2013), the distribution of ARGs in geography remote locality might be due to transmission through airborne bacteria and migratory birds.

Metagenomic study discloses the presence of various antibiotic resistance genes belonging to quinolone, bacitracin, tetracycline, beta-lactam, aminoglycoside. These genes also showed the maximum identity with Gram-negative bacteria and psychrotolerant bacteria. For example, the gene related to beta-lactam was showing identity with *Serratia marcescens* strain SR50 and tetracycline was showing identity with *Bifidobacterium bifidum* strain NCIMB 4117. Similarly, all the other predicted genes showed in functional metagenomic results were having close similarity with psychrotolerant Gram-negative bacteria and many similar types of bacteria were also detected through metagenomic analysis (Segawa et al., 2013). From various glaciers of the world, diverse ARGs were reported such as quinoline resistance genes were detected from Yala glacier (Nepal), Qiye glacier (China), and Gulkund glacier (Alaska). Similarly, beta-lactam resistance genes were recovered from Yala glacier, Qiye glacier, Takeyama glacier (Japan) and genes related to tetracycline resistance were also reported from Yala glacier, Qiye glacier, and Takeyama glacier.

Thus, it can be concluded that the resistance gene detected through metagenomics might be biologically transmitted from microorganisms through lateral or horizontal gene transfer, in particular micro-environments where dense microbial communities are often exposed to an intensive use of antibiotics (Segawa et al., 2013). Many researchers reported that MRGs are close linked to antibiotic resistance phenomenon (Pal, 2017). Environmental metal resources for copper and zinc heavy metals are abundantly present. Since, early days these heavy metals have been correlated and connected to the increase of heavy metal tolerance in environmental microflora (Seiler and Berendonk, 2012; Wales and Davies, 2015). In ecological samples, copper and zinc have always been found along with the related presence of antibiotic-resistant microbial communities (Holman and Chenier, 2015). According to Hu et al. (2016) the chances of transmitting mobile

genetic elements (MGEs) increases with increase in the concentration of copper.

There are several reports on metals and biocides as promoters of antibiotic resistance via co-resistance. For instance, the resistance gene to copper (*ctrB*) has been linked with resistance genes against erythromycin, vancomycin, and tetracycline by Amachawadi et al. (2011) and Hasman and Aarestrup (2002). The genes *oqxAB* coding for efflux pump, confers resistance to triclosan, acriflavine, chlorhexidine, and fluoroquinolones, were shown to be co-located with beta-lactamases (*blaCTX-M*), copper (*pco*) and silver (*sil*) operons on the same plasmid, therefore demonstrating their potential for co-selection (Fang et al., 2016). The MRGs were closely correlated between the antibiotic-resistant phenomena by many researchers (Seiler and Berendonk, 2012). Thus, this led us to hunt for MRGs in a glacial environment.

The copper and zinc resistance gene elements have always been detected parallelly with the antibiotic-resistant microbial communities in the environmental samples (Peltier et al., 2010; Becerra-Castro et al., 2015). The chances of transfer of mobile genetic elements increases with an increase in the concentration of copper, acquiring the ability of copper-dependent ARGs that are readily accessible (Dupont et al., 2011). Antibiotics also form strong metallo-ligand coordination complexes with metal ions and their ionic interaction can illustrate broad results (Niebergall et al., 1966). The inactivation of penicillin (Cressman et al., 1966) by zinc was first reported by Eisner through β -lactam hydrolysis (Eisner and Porzecanski, 1946). Thus, we primarily screened for the metal tolerance (if present) among our bacterial isolates from the glaciers. Fascinatingly, it was found that many of our bacterial isolates were tolerant to heavy metals, demonstrating higher MICs than *E. coli* (Nies, 1999).

The functional metagenomic analysis also showed the presence of heavy metal resistance genes in glacial samples as discussed earlier. The MRGs also showed maximum identity, i.e., more than 97% with psychrotolerant bacteria. The copper resistance genes showed (96%) identity with *Pseudomonas fluorescens* and *Escherichia coli* K12; whereas mercury resistance genes (*merA*) showed (98%) identity with *Thiobacillus ferrooxidans*, *Staphylococcus aureus*, *Pseudomonas stutzeri*, and *Pseudomonas aeruginosa*. The chromium resistance gene (*chrA*) showed (96%) identity with *Pseudomonas aeruginosa* and arsenic resistance genes (*arsH*) showed (98%) identity with *Bacillus subtilis* strain 168. Arsenic resistance genes were detected only at Changme Khangpu glacier. MRGs for copper and chromium were identified from debris-covered glacier Changme Khangpu glacier, whereas Changme Khang glacier which was

debris free glacier had MRGs for cobalt and iron. This interesting observation led for Inductive Couple Mass spectroscopy (ICP-MS) analysis, where it was found that the elemental composition of both the glaciers were almost similar. The pH and the concentration of copper and iron were similar in both the glaciers (Sherpa et al., 2018b). The only difference was that the Changme Khang glacier had higher percentages of calcium and magnesium concentration in comparison to Changme Khangpu glacier (Sherpa et al., 2018b). Therefore, it can be assumed that the difference in MRGs was due to the microbial composition of two glaciers and not due to the elemental composition of these glaciers. In our earlier study, we found that the bacterial community composition of these two glaciers showed variation. Among the two glaciers, Changme Khang (54.04%) had higher percentage of Gram-positive bacteria than Changme Khangpu (24.84%), whereas, Gram-negative bacteria were higher in Changme Khangpu (22.65%) than Changme Khang (4.41%) (Sherpa et al., 2018b). Although at the phylum level diversity, Proteobacteria was dominant followed by Firmicutes, Actinobacteria, and Ascomycota; however, species level diversity varied significantly among the two glaciers (Sherpa et al., 2018b). In case of Changme Khang glacier, the major species detected were *Serratia marcescens* (31.19%), *Delftia acidovorans* (19.79%), *Stenotrophomonas maltophilia* (2.26%), *Pseudomonas putida* (0.51%) and *Escherichia coli* (0.44%); whereas in the case of Changme Khangpu glacier, it was dominated by *Delftia acidovorans* (37.59%), *Serratia marcescens* (24.80%) *Stenotrophomonas maltophilia* (0.86%). *Bacillus cereus* and *Bacillus thuringiensis* together accounted for about 0.41% of the total sequences, and *Pseudomonas putida* accounted for 0.11% of the total sequences respectively (Sherpa et al., 2018b). The various metal resistant genes detected by functional metagenomics were related to genus *Pseudomonas* and *Bacillus*. These bacteria were also detected by metagenomics and culture dependent study of Changme Khang, Changme Khangpu glacier and Kanchengayo glacier, North Sikkim, India (Sherpa et al., 2018b).

Thus, our results supported the theory that the antibiotic and heavy metal resistance genes detected through metagenomic may be co-occurrence of heavy metals resistance to antibiotic resistance. Therefore, it might be suggested that there might be co-occurrence or co-selection of these genes in such environments. Hence, this is a broad hypothesis which needs to get counter to verify with other cold-habitats.

5. Conclusion

Interestingly, we found that majority of the bacterial isolates were resistance against the most of antibiotics tested against. These results were supported by metagenomic analysis. Metagenomic study disclosed the presence of various antibiotic resistance genes belonging to quinolone, bacitracin, tetracycline, beta-lactam, aminoglycoside. These genes showed the maximum identity with Gram-negative bacteria and psychrotolerant bacteria. The isolates have shown substantial tolerance towards five heavy metals as compared to *E. coli*. Also, the different types of metal resistant genes were detected by metagenomic analysis. Thus our isolates which are resistant to some antibiotics and also tolerant to heavy metals support the hypothesis that there might be co-occurrence or co-selection of antibiotics resistant genes and metal resistance genes in such environments, which was further supported by detection of different kinds of metal resistance genes by metagenomic analysis. However, this is an inconclusive assumption which needs further research in such habitats.

Credit author statement

Mingma Thundu Sherpa: Conceptualization, Investigation, Methodology, Writing-Original draft preparation, data compilation, Formal analysis. **Ishfaq Nabi Najar:** Investigation. **Sayak Das:** Investigation. **Nagendra Thakur:** Writing, Review, Editing, and Supervision.

Declaration of competing interest

Authors have no conflict of Interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoenv.2020.111037>.

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